

3. On the Invariant Theory of Forms in n Variables.^[101:1]

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In the projective invariant theory of forms in n variables, the main problem has been solved: the finiteness of the system of forms has been proved. A number of further questions, however, have not been treated — namely, those concerning methods for studying the interrelations among the forms. I should like to report briefly the results I have obtained on such questions, but, for clarity, I shall first outline the questions in the ternary domain, where they are familiar.

I have in mind first the two fundamental theorems of the symbolic method: the theorem that all invariant constructions can be represented symbolically, and the second theorem that all relations among invariants are obtained by successive application of a small number of symbolic identities; thus the symbolic method gives all relations without leaving the domain of invariants.^[102:1] Upon these are based the processes for generating forms: the symbolic folding process and the nonsymbolic differentiation processes, the theory of series expansions, and the like. In this connection I should also mention a theorem proved in my dissertation,^[102:2] namely, that all foldings can be generated by successive application of the two possible “cogredient” foldings; for a symbolic product $s_x t_x u_\sigma u_\tau$, by these I mean the two foldings (stu) and $(\sigma\tau x)$. The theory of reducers and the reduction of systems of forms can be built on this theorem, analogously to the binary domain, as I showed in my dissertation.

When we now pass to the n -ary domain, even the first theorem of the symbolic method holds only with qualifications. As is well known, for quaternary forms there already occur, besides the point and plane coordinates x and u , line coordinates p_{ik} , the minors formed from two rows of point coordinates. Analogously, in the n -ary domain we have variable rows p_ρ , whose individual elements are the minors formed from ρ rows x , and symbol rows S^ρ , whose elements behave like minors formed from ρ rows s cogredient to the u . The symbolic representation by determinants is known to hold only when the symbol rows S^ρ are resolved into the rows s and these are distributed among different determinants; only aggregates of determinants that depend on the S^ρ then have genuine significance. There is no systematic way to tell which determinant aggregates have this property, and all the other theorems mentioned in the ternary domain are consequently lost.

I should now like to give a simple principle for obtaining a *symbolic representation explicit in the rows S^ρ* , and thereby also recovering the remaining theorems. This is an application of the familiar theorem on corresponding matrices: “Every variable row p_ρ is in one-to-one correspondence with another row $q_{n-\rho}$, whose individual elements are the minors formed from $n - \rho$ rows u connected with the ρ rows x by equations $(u | x) = 0$.” Correspondingly, to every symbol row S^ρ there is associated another row $S_{n-\rho}$, which behaves as though composed of $n - \rho$ rows cogredient to the x .

The theorem admits a second formulation suited to applications: “To every matrix product $(v^{(1)} \dots v^{(\rho)} | y^{(1)} \dots y^{(\rho)})$,^[103:1] where the rows of the second matrix are contragredient to those of the first, there corresponds a determinant; and conversely every determinant can be transformed into a matrix product with a prescribed number ρ of rows.” Determinant and matrix product thus appear completely equivalent, and the first theorem of the symbolic method takes the form: “All invariant constructions can be represented symbolically by matrix products.” From two symbol

^[101:1]Lecture delivered at the Salzburg Meeting of Natural Scientists, 1909.

^[102:1]Study: *Methoden zur Theorie der ternären Formen*. II. § 6. (Leipzig, Teubner 1889.)

^[102:2]Über die Bildung des Formensystems der ternären biquadratischen Form. § 1. (Crelle’s Journal Bd. 134. 1908.)

^[103:1]That is, the sum of the products of corresponding determinants formed from the matrix of the ρ rows v and the matrix of the ρ rows y .

rows S^σ, T^τ , with the aid of variable rows, it is now very easy to form a matrix product *explicitly* containing these symbol rows and therefore representing an invariant construction of the form $(S^\sigma | p_\sigma)(T^\tau | p_\tau)$, namely:

$$(1) \quad (q_{n-\sigma-\lambda} S^\sigma | T_{n-\tau} p_{\tau-\lambda}), \quad (\lambda = 1, 2, \dots, \tau \text{ or } n - \sigma).$$

More important is the converse: *that all invariant constructions can be represented explicitly by matrix products*; hence the foregoing process is the extension of the binary and ternary folding process. To prove this converse, it is necessary to transfer the second theorem of the symbolic method to the n -ary domain, namely, to set up the complete set of independent indecomposable identities in which all rows S^ρ, p_ρ are treated as indecomposable. These identities follow from the familiar ones involving only rows x and u ^[103:2] by replacing the product theorem for determinants with a system of product theorems for matrices, and by combining different matrix products into a single one by means of those same product theorems. In this way one obtains two dual formulas, of which only one will be stated:

$$(2) \quad (S_1^{\rho_1} S_2^{\rho_2} \cdots S_k^{\rho_k} | x p_{\rho-1}) = \sum_{i=1}^k \varepsilon (S_1^{\rho_1} \cdots S_{i-1}^{\rho_{i-1}} S_{i+1}^{\rho_{i+1}} \cdots S_k^{\rho_k} q_{n-\rho+1}, S_{n-\rho_i} x);$$

$$\rho = \sum_{i=1}^k \rho_i \leq n; \quad \varepsilon = \pm 1.$$

The only specialization contained in these formulas — the occurrence of a row x — cannot be avoided; with completely general symbol and variable rows we arrive at a “decomposition identity,” an identity in which a row p_ρ is decomposed into the individual rows x . Clebsch already used the simplest special case of the decomposition identity in his “fundamental problem of invariant theory” to derive the elementary covariants in the theory of series expansions. With the aid of these identities, which mediate the transition from the non-explicit determinant expressions to the matrix product, it can indeed be shown that the matrix representation yields the desired explicit symbolic representation.

For the folding process, however, a theorem exactly analogous to that in the ternary domain holds, and the reduction theorems can likewise be based on it. If in (1) the number λ is called the defect of the folding, and foldings for which $\sigma = \tau$ are called cogredient, *then all foldings can be generated by successive application of cogredient foldings of defect 1*. The proof of this theorem rests essentially on the fact that the most general forms can be replaced by “normal forms,” that is, by forms that vanish identically under every invariant differential operation. In the quaternary domain, this latter fact was proved by Mertens;^[104:1] our knowledge of the most general differential operations — which, as is well known, arise from the folding processes by replacing the symbol rows with differential symbols — permits us, using the decomposition identity, to carry Mertens’s proof over almost word for word to the n -ary domain.

^[103:2]E. Pascal in *Memorie d. R. Acc. d. Lincei*. (4) 5 (1888), p. 375.

^[104:1]Über invariante Gebilde quaternärer Formen. Wiener Berichte. Bd. 98. 1889.